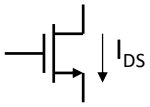
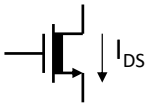
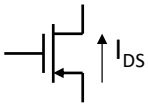
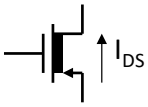
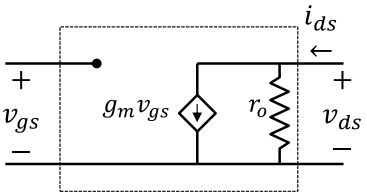
COGNOME _____ NOME_____ MATRICOLA_____

NMOSFET		PMOSFET	
			
E-nMOSFET $V_{TN} > 0$	D-nMOSFET $V_{TN} < 0$	E-pMOSFET $V_{TP} < 0$	D-pMOSFET $V_{TP} > 0$
Zona di saturazione			
$I_{DS} = \frac{k_n}{2}(V_{GS} - V_{TN})^2$ Condizioni: $V_{GS} > V_{TN}$ e $V_{DS} > V_{GS} - V_{TN}$		$I_{DS} = \frac{k_p}{2}(V_{GS} - V_{TP})^2$ Condizioni: $V_{GS} < V_{TP}$ e $V_{DS} < V_{GS} - V_{TP}$	
Zona lineare o triodo			
$I_{DS} = k_n\left(V_{GS} - V_{TN} - \frac{V_{DS}}{2}\right)V_{DS}$ Condizioni: $V_{GS} > V_{TN}$ e $V_{DS} < V_{GS} - V_{TN}$		$I_{DS} = k_p\left(V_{GS} - V_{TP} - \frac{V_{DS}}{2}\right)V_{DS}$ Condizioni: $V_{GS} < V_{TP}$ e $V_{DS} > V_{GS} - V_{TP}$	
Zona di interdizione			
$I_{DS} = 0$ Condizioni: $V_{GS} < V_{TN}$		$I_{DS} = 0$ Condizioni: $V_{GS} > V_{TP}$	
Modello ai piccoli segnali (in saturazione)			
			
$g_m = k_n(V_{GS} - V_{TN})$ $r_o = \frac{2}{k_n(V_{GS} - V_{TN})^2\lambda_n}$		$g_m = -k_p(V_{GS} - V_{TP})$ $r_o = \frac{2}{k_p(V_{GS} - V_{TP})^2\lambda_p}$	